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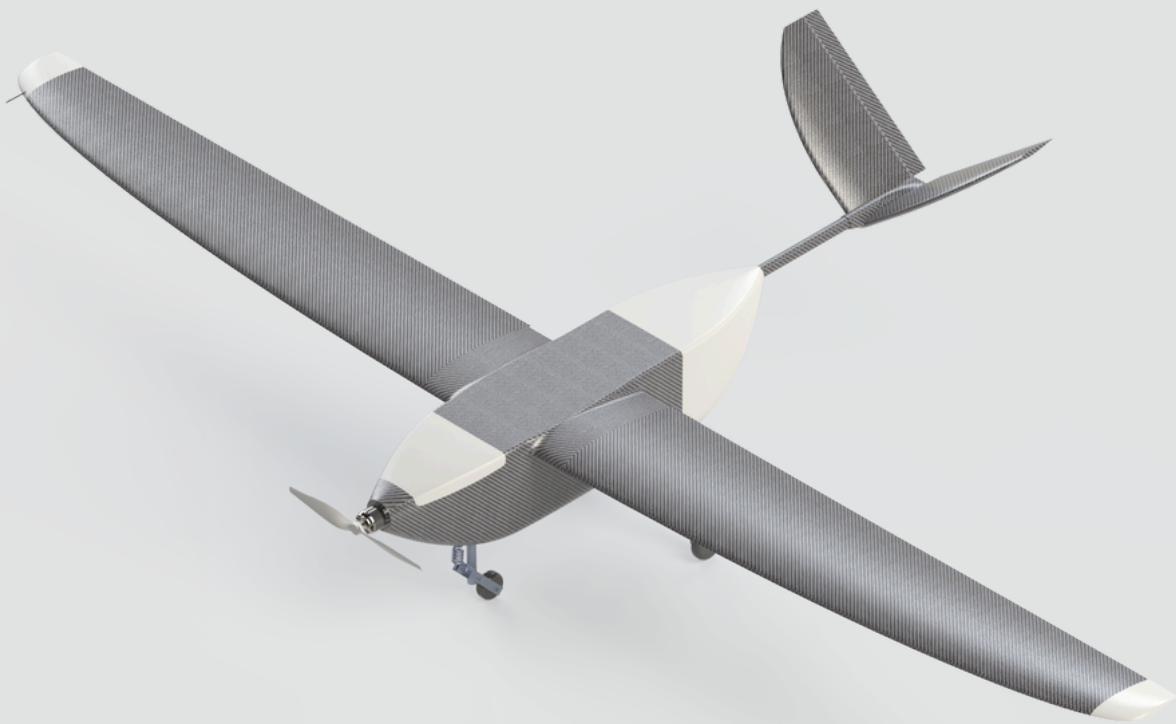
DEPARTMENT
OF AEROSPACE SCIENCE
AND TECHNOLOGY

FLY MI

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DESIGN & DEVELOPMENT SPECIFICATION

13 - Fly-Mi - Politecnico di Milano



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1 Design Summary

O.L.I.V.I.A. (Optimized Lightweight Intelligent Vehicle for Immediate Aid) is a fixed-wing unmanned aerial vehicle (UAV) designed for the delivery of humanitarian aid in crisis situations. Its primary role is to provide logistical support in disaster-affected regions, such as those impacted by hurricanes or severe flooding. In such scenarios, affected areas may become isolated, necessitating an effective means of transporting essential supplies, including food and medical aid, to remote or hard-to-reach locations.

Configuration Design As shown in the cover image, the UAV features a traditional longitudinal configuration with an aft-tail and a high-wing design, optimised to accommodate a jettisonable parachute-equipped payload within the cargo bay. Housing the payload within the fuselage maximises lift-to-drag ratio, thus enhancing total operational range of the system. This is critical in missions where range of operation is highly variable, as extended range capability ensures access to more remote locations. To further enhance aerodynamic efficiency, a V-tail configuration has been selected. This design not only improves aerodynamic performance but also contributes to weight reduction without significantly compromising flight stability and control characteristics. The propulsion system is fully electric, operating exclusively on quick-interchangeable battery modules. This approach maximises operational efficiency and on-ground versatility. Finally, a tricycle landing gear configuration has been implemented due to its versatility in take-off and landing operations. This configuration enables it to operate effectively on unprepared runways, enhances ground stability, and reduces both take-off and landing distances. These characteristics are essential to meet competition requirements and ensure reliable operation in unknown environments and challenging terrains.

Compared to what stated in the **Concept Paper, PDR and CDR**, the design has evolved remaining almost the same, the major changes are listed in the table below.

Applied Change	Reason for Change
Introduction of a wing twist of -2° (mid-to-tip)	Initially designed with zero wing twist, which was introduced to improve the lift-to-drag ratio and enhance stall characteristics.
Steerable mechanism instead of a fixed front wheel	Required for the <i>ArduPilot</i> software to actively and optimally control the UAV on the ground during autonomous take-off.
Payload Release System dimensions	Payload bay dimensions changed in order to accommodate the new dimensions of the <i>AirDropBox</i> , making sure to have room for the full length and correct deployment.
Propeller dimensions	The propeller dimensions have been identified after the propulsion tests and calculations.

Table 1: Design modifications.

Payload Capability A detailed analysis of the design, aircraft balance, and manufacturing process was continuously updated throughout each phase of the project, ensuring an accurate weight estimation. This approach allowed for effective monitoring of the payload mass and the corresponding competition score during the Fly-Off event. A comprehensive breakdown of the weight calculations is provided in the Design Description section. The estimated payload mass is reported as: $M_{PL} = 2.5$ kg.

Sponsor Engagement The support of the sponsors was fundamental to O.L.I.V.I.A.'s in providing advice, advanced materials, specialized equipment and technologies. The table below summarises sponsor's contributions to the project.

Sponsor	Contribution	Sponsor	Contribution
<i>Bercella</i>	Wing moulds	<i>CIT</i>	Composite materials
<i>Caracol</i>	Fuselage moulds	<i>Fiber Compositi</i>	Composite materials
<i>Maprint 3D</i>	Tail moulds	<i>Sorvolando Compositi</i>	Composite materials
<i>Euromodel</i>	Landing gear moulds	<i>Dassault Systemes</i>	Software
<i>Aerosviluppi</i>	Project Management consulting	<i>Ansys</i>	Software
<i>Eligio Re Fraschini</i>	Workspace and autoclave	<i>MKS</i>	Servos
<i>SIR Safety System</i>	Personal Protective Equipment	<i>Xenia, Guztex Polymers</i>	Mould materials

Table 2: Sponsor contributions.

2 Project Management

2.1 Overall Approach

The O.L.I.V.I.A. project was developed through several key phases. The starting point was a thorough analysis of the previous year's project, aiming to retain effective approaches while improving areas that had encountered challenges. Based on this analysis, an initial Gantt chart¹ was developed as a foundation for the final version. Compared to the previous year, a more practical approach was adopted, prioritizing testing phases as much as possible. Additionally, a delay forecast was conducted, taking into account critical periods such as exam sessions.

Given the fundamental importance of preliminary planning, we sought support from both technical sponsors and partners for project management consulting. Through these consultations, we refined our planning process by creating a Work Breakdown Structure (WBS), outlining the key steps for each department, including dependencies and critical paths. Alongside this, we conducted a resource allocation study to optimise workload distribution within the association. As a result, we established a final Gantt chart which has been continuously updated to reflect unforeseen events and remains the central reference for project execution.

Once the Gantt chart was finalized, the Business Management & Communication department took action, developing a project budget, analyzing financial risks, securing sponsorships, and coordinating with external entities involved in critical activities to minimize potential delays caused by external dependencies.

In parallel, an analysis of the scores achieved by last year's competitors was conducted and cross-referenced with this year's scoring methodology. This allowed the team to develop a winning strategy and identify the key activities necessary to maximize the final score.

2.2 Progress Review & Next steps

The project is progressing on schedule, with no major delays affecting key milestones. The design phase was successfully completed, as documented in the CDR and PDR reports. In parallel, the team has conducted several in-depth studies to refine the design:

- Comprehensive testing campaign on the **propulsive system**, optimizing propeller and battery selection.
- Advanced **CFD RANS simulations** conducted by the aerodynamics team to evaluate O.L.I.V.I.A.'s configuration, providing insights for trade-off analysis and performance assessment.
- Continuous updates to the **weight breakdown** by the structures department to reflect modifications made during design reviews.

Beyond design refinements, significant efforts have been dedicated to the lamination process, ensuring that all necessary moulds, workspaces, and materials are ready on schedule. This required extensive coordination with sponsors and careful logistics planning for material transportation. However, a delay in the fuselage mould milling, caused by sponsor availability, necessitated an adjustment in the manufacturing sequence. To mitigate this, wing lamination and assembly were prioritised, maintaining steady progress.

A key amendment to the initial project plan involves the second flight testing campaign. Originally, Nyx, the 2023-2024 drone, was intended as the testbed, however, restoring and modifying Nyx would require extensive repairs and system upgrades, exceeding the team's resource availability and making it unfeasible within the predefined schedule. Consequently, autonomous system testing will continue using Fabia, the drone used in the first testing campaign.

To further enhance efficiency, a specialized task force has been formed to focus exclusively on the development of the X-Plane model. This ensures high-quality work while minimizing task dispersion in one of the competition's key evaluation areas.

¹For space reasons, the Gantt chart is not included here. For more information, see the Gantt chart in the concept paper.

Remaining Project Steps The table below outlines the remaining key activities, their expected duration, responsible teams, and dependencies.

Task	Duration (weeks)	Responsibility	Dependencies
Component lamination	2	Manufacturing Team	Sponsor curing process
Wing Assembly	1	Assembly Team	Lamination completion
Tail Assembly	1	Assembly Team	Lamination completion
Control Surfaces Assembly	1	Assembly Team	Lamination completion
Fuselage Mold Milling	2	Sponsor	External delay
Fuselage Lamination	2	Structures dep.	Mold availability
Final Assembly	2	Assembly Team	Completion of all components
Avionics Integration	1	EPC (Electronics Propulsion & Control dep.)	Final assembly completion
Ground Tests	1	Testing	Avionics and structural completion
Flight Readiness Review	2	EPC	Ground tests completion
Software Testing Campaign	4	EPC	System updates, checklists, procedures
X-Plane Sim. Development	3	Simulation Task Force	Software tools, prior data
Flight Testing Campaign	4	EPC	Flight readiness review
Competition preparation	2	Competition Team	Flight testing completion

Table 3: Remaining project steps.

Despite minor delays, the team has successfully adapted by strategically prioritizing tasks. By maintaining a structured workflow and leveraging specialized task forces, the project remains on track for successful completion within the planned timeline.

2.3 Project Risk Assessment and Mitigation Measures

To ensure project success, a thorough risk assessment was conducted across key areas such as resourcing, skills, procurement, manufacturing, and testing. Early risk identification enables proactive mitigation, minimising disruptions, and ensuring smooth execution. The table below presents key risks, their explanations, and corresponding mitigation measures.

Category	Type of Risk	Explanation	Mitigation
Skills	Lack of members with experience and specific skills	Lack of qualified members leads to design errors, operational inefficiencies, and project delays. High turnover makes knowledge transfer difficult.	Organization of workshops, collaboration with professors and specialized companies, creation of technical documentation to pass on to future team members.
Coordination	Silo effect	Lack of communication between different technical departments can cause isolated workflows and lead to inefficiencies, duplicated efforts, and integration issues.	Communication encouragement, shared materials and meeting minutes accessible, two department heads for mutual oversight, check-up forms for all members after each major project phase.
Resourcing	Lack of funds for materials and equipment	The scarcity of financial resources can cause development delays or force less efficient technical solutions.	Expansion of the sponsor network, participation in university grants, fundraising, and collaboration with supplier companies.
Procurement	Delays in materials and components supply	Dependence on external suppliers can lead to delays in assembly and testing times.	Advance planning of purchases, agreements with multiple suppliers, creation of minimum stock and time buffers.
Design	Calculation errors in the drone design phase	Errors in aerodynamic, structural, and electronic calculations can compromise the drone's stability and efficiency.	Comparison of results with industry experts and professors conducting tests.
	Incorrect choice of materials to use	Using unsuitable materials can affect weight, strength, and durability.	Consultation with experts, use of analysis software, comparative tests with different materials.
Manufacturing	Assembly & lamination issues	Errors in assembly and lamination can cause malfunctions, waste of materials and require redesign.	Creation of detailed guides and checklists, progressive tests, and constant review of component compatibility.
Flight Testing	Technical failures during tests	Flight tests may reveal problems with avionics, motors, or structure, causing damage and delays.	A safety analysis has been carried out as described in the Safety section of this report. Implementation of safety checklists.

3 Requirement Review

Structural and Construction Requirement Review Table			
Requirement	ID	Actions Taken	Remaining Measures
Maximum Take Off Mass (MTOM) under 10.0 kg	4.1.2	Optimized structure using lightweight materials and iterative design to reach a 7 kg target, identified via the Size Matrix Plot (SMP) method. 20% margin added to component mass estimates for unforeseen increases, with the final weight validated by direct measurement.	Total weight of the UAS verification after lamination and assembly completion.
Storage container - external dimensions of 1500 mm x 600 mm x 600 mm	4.5.1	A CAD model has been created for box design and positioning of parts within box dimensions has been studied. Compartment design for easy unloading and protection of drone parts.	Construction and measurements verification.
No tools routine assembly/disassembly	4.5.2	Connections between wing, fuselage, and tail designed for tool-free assembly. The wing-fuselage uses a tube and pin, with horizontal translation constrained by drawbolts. Tail is bonded to the boom and slides through the fuselage, secured by bulkheads and a spar.	
Payload of at least 1.25 kg of sand	4.4.1	Structure designed to minimize structural weight and maximize payload weight. Payload currently estimated at 2.5 kg.	
Secondary retention mechanisms	4.5.3	Secondary retention mechanisms have been provided for the major sub-components.	
Retention straps (payload) must be releasable without tools	4.4.2	Pin holder on tail boom, with a pin activated by a servo motor. Payload secured with cables, one end attached to the pin and the other to the tail boom. After release, cables are stored in the fuselage. No tools needed.	
Tracker with clear view of the sky, 140-degree field of view	4.4.3	Fibreglass and rohacell cover designed in the rear fuselage for aerodynamic reasons used to support electronic elements that cause issues in contact with carbon fibre. Balsa wood shelves ensure right positioning.	Construction and design validation.
Vehicle designed to carry an RF locator	4.4.4	Fibreglass and rohacell cover used to support electronic elements such as the RF locator.	
Batteries positively secured	4.2.1	Removable two-compartment shelves designed to accommodate all the electronics. Batteries will be secured to the upper shelf.	Validation in non-destructive tests.
Wiring retained securely	C.4	Wires placed in ordered way during construction, so that they will not run near critical spots or mechanisms or in Flight Controls proximity.	Cable positioning tests.
Batteries protected in case of a crash	4.2.1	Battery located on the upper and safest part of electronic bay, close to the faring and therefore easy and quick to remove by hands.	
Electronics Propulsion and Control Requirement Review Table			
Automatically navigating via GPS coordinates	4.1.7	Satisfied by Ardupilot on a testing drone.	Validation in testing campaign.
UAS capable of storing Geo-fence coordinates	4.1.7	Satisfied by Ardupilot on a testing drone.	Validation in testing campaign.
Autonomous command signal to activate FTS if breaking Geo-fence limits	4.1.7	Satisfied by Ardupilot on a testing drone, using a FTS sequence which did not destroy the plane on activation.	Same test on the final UAS in testing campaign.
15 minutes flight with 20% battery capacity left.	A.3.2	Tested through simulation and validated in testing campaign.	
Audible and visual alert	4.1.10	Satisfied by Ardupilot on a testing drone.	Validation in testing campaign.
Only batteries for energy storage	4.2.1	One LiPo 6S battery have been selected to be the only energy storage device on the aircraft.	
Lithium Polymer (LiPo) batteries	4.2.1	One LiPo 6S battery has been selected to be the only energy storage device on the aircraft.	
Bright colours in batteries	4.2.1	The batteries will have some colourful tape on each side in order to be easily visible also in scarce-visibility environment.	
Clearly labelled link to isolate power to the UAS	4.2.3	The safety switch to the aircraft placed on the bottom of the aircraft, away from the propeller, easily accessible and labelled as specified.	
Flown manually or automatically during the whole mission	4.3.1	Flight carried on by Ardupilot in autonomous mode, from take-off to landing. The possibility to take the manual control when necessary is guaranteed.	Validation in testing campaign.
Revert to manual control at any point	4.3.1	Manual control and FTS set on radio switches. The functionality of these switches will be tested before each flight.	
Controller in Mode 2 configuration	4.3.1	Radio will be set and kept on Mode 2. This will be verified before each flight.	
FTS sequence	4.3.2	FTS correctly programmed in Ardupilot to execute required sequence, eventually through the use of Lua scripts. No parachute designed. Fail-safe validated before each flight.	Non-destructive on-ground and in-flight tests.

FTS trigger-events	4.3.2	FTS triggered by geo-fence breach, signal loss, datalink loss. It will be possible to manually trigger FTS from both radio and GCS.	All trigger-events testing validation.
GCS information	4.3.3	GCS with 2.4GHz link separated from radio link, ensuring safety and redundancy in case of radio link loss. GCS connected to a PC, displays telemetry and is capable of controlling the aircraft (and of triggering FTS). PC connected to a tablet, to have a more portable device near pilot. Radio itself capable of showing the telemetry.	
Suitable port for wired connection between air vehicle and GCS	4.3.3	USB interface + buzzer present on flight controller and easily accessible on the plane. A USB cable will always be present for PC connection.	
Radio equipment, ranges of at least 1 km	C.5	2.4GHz true-diversity RC receiver and radio run ExpressLRS, ensuring one of the best performances on the market in terms of signal quality and range (validated in testing). Two antennas will be placed on the bottom of the plane, outside the fuselage, one for each side, ensuring full coverage during the mission.	
‘Spread Spectrum’ 2.4 GHz band	4.3.4	2.4GHz true-diversity RC receiver has been chosen.	
Electric motor	C.1	Selected propulsion system includes a T-Motor AT4120 KV560, which satisfies the requirement.	
Geo-Fence with margin to allow for drift	C.8	Simulation of flight path carried on Mission Planner before each flight, applying necessary corrections. Path shall never go too close to the geo-fence. A safety margin will be considered on top of the simulation.	
Aid Package deployed by automatic command	4.4.2	Payload Release system is designed to be activated by an automatic command.	PRS activation and parachute compliance verification.
Financial Requirement Review Table			
COTS costs under £1300	4.7	COTS spreadsheet kept updated to ensure compliance with the demand.	
Secondary Requirement Review Table			
Empty CG marked	4.1.2	Precise CG position calculated.	Manual CG verification on final UAS.
Bright colours	4.1.10	Bright colours livery under development.	Livery materials identification.
Operational Requirement Review Table			
Take-off in 10 m x 10 m box	4.1.6	Take-off requirement considered in the Size Matrix Plot. The design point found satisfies the demand.	
Landing in 10 m x 20 m box	4.1.6	Flaps designed with the AER department in compliance with the landing requirements.	
Max 200 ft AGL	4.6.5	Geo-fence on altitude imposed on Ardupilot, telemetry will always kept under pilot's sight.	
Max 40 kts true airspeed (KTAS)	4.1.4	Maximum airspeed imposed on Ardupilot. Ardupilot speed limit functionality validated on a testing drone.	Validation in UAS testing campaign.
Max bank angle around 30 degrees	C.8	Maximum bank angle imposed on Ardupilot, as well as navigation parameters tuning in accordance to the UAS capabilities.	
Winds up to 20 kts gusting to 25 kts, light rain	4.1.5	V-tail with a low butterfly angle (38°) guarantees low wind sensibility and great static and dynamic stability.	
5 kts (8 kts gusts) cross-wind take-off and landing	4.1.5	vertical projection of V-tail minimized while maintaining good stability parameters, minimizing UAS susceptibility to gusts of wind.	
UAS capable of flying again with fresh battery and no repairs	4.1.6	tail strike during take-off and landing considered in tail boom sizing, preventing tail impact during take-off rotations and landing flares.	
Hard or grass runways	4.1.6	Runway type considered in landing gear design, considering past years models experience.	
In-flight payload release	4.4.2	Payload release system designed to be activated only during flight.	
UAS assembly and preparation within 6 minutes	5.3.5	Aircraft's structure and joints designed to be as quick and intuitive as possible.	Preparation time testing on final UAS.
Catapults setting up	5.3.2	No catapults needed.	
Take-off preparation in 5 minutes	A.2.4	Easy-to-assemble structure, checklists for pre-flight checks.	
Max 15 minutes of flight	A.3.2	Compliance with this requirement tested through simulation.	Validation during final UAS testing campaign.
Flight within VLOS from the Pilot	4.6.5	Mission designed within the pilot's VLOS. In case of aircraft loss, FTS will be activated in accordance with the corresponding requirements.	

4 Design Description

4.1 Flight Mechanics

The Flight Mechanics department is responsible for the preliminary sizing of the aircraft, including the design of the tail, the sizing of the control surfaces, the evaluation of the aircraft balance and the analysis of performance and stability characteristics. For this project, the main focus was on designing an aircraft that could provide great stability and easy control during every phase of the mission, even with adverse wind conditions and the difficulties associated with an autonomous flight system. At the same time, there was a strong collaboration with other departments to ensure the correct implementation of critical design features, such as the centre of gravity positioning, with and without payload, and the optimisation of important components, such as the weight and efficiency of the tail and its boom.

V-tail Motivations Following an analysis of top-performing teams and a review of literature on tail configurations, the V-tail was chosen for O.L.I.V.I.A. due to its multiple advantages. It simplifies manufacturing and reduces overall weight while also minimising interference drag, enhancing aerodynamic lift-to-drag ratio. Additionally, it provides sufficient stability and controllability, ensuring reliable flight performance. The reduction of downwash effects further improves efficiency compared to traditional tail designs.

Tail Optimization Following tail shape and configuration selection, certain parameters required further refinement, including butterfly angle, surface area, and boom length. The objective was to minimise weight while ensuring adequate stability characteristics. To achieve this, an optimisation algorithm was developed in MATLAB to evaluate multiple configurations and determine the optimal solution. Subsequent post-processing allowed for the refinement of the tail dimensions and butterfly angle, resulting in overall weight reduction and aerodynamic lift-to-drag ratio improvement. The results of this study were found to be **H-tail volume coefficient** = 0.514, **V-tail volume coefficient** = 0.017, **Dihedral angle** = 38° , **Boom length** = 1.05 m from wing root's leading edge.

Trim and Performance Analysis

The entire aircraft was modelled in XFLR5 to study its trim condition, its performance characteristics and its stability, control parameters and to check the aircraft balance. The team aimed to design a plane trimmed in a condition as close as possible to the maximum lift-to-drag ratio attitude, which corresponds to the *best range* condition. To do so, some parameters have been tailored to meet this request, such: **Static margin** = 10%, **Tail incidence** = -2.2° .

The balance was continuously checked using a lumped-mass model and the CG position calculated throughout the whole development and the static margin ensured. The latest check was performed using the masses found in Table 8 using the XFRL5 software tool.

Following the trim analysis and the detailed aerodynamic polar study presented in the Aerodynamics section, the performance characteristics of O.L.I.V.I.A. were evaluated and verified. These include the true aerodynamic lift-to-drag ratio, the corresponding best range velocity (V_{BR}), the thrust requirements for take-off, and an estimate of the optimal cruise speeds. It is worth noting that the cruise speed, corresponding to the maximum lift-to-drag ratio, is close to the stall speed (clean configuration). This proximity posed a potential safety risk when operating under an autonomous system such as Ardupilot. To mitigate this, a safety margin of 2.5 m/s was added to the cruise velocity. Finally, the final key performance parameters have been calculated using models found in the literature and the whole mission scenario has been simulated in the said condition and can be summarised in Table 4, which shows all flight phases, including both the two laps of the cruise circuit with payload and the two without.

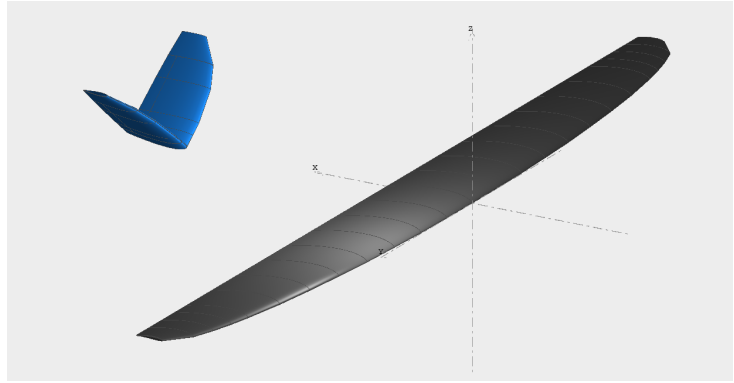


Figure 1: XFLR5 model of O.L.I.V.I.A.

- **Take-off thrust:** $T_{TO} \simeq 60$ N
- **Take-off distance:** $s_{TO} \simeq 10$ m
- **Cruise speed (PL):** $V_{cruise}^{PL} = 15.5$ m/s
- **Cruise speed (no PL):** $V_{cruise}^{no PL} = 13.0$ m/s
- **Stall speed (landing):** $V_{stall}^{LND} = 8.5$ m/s
- **Landing distance:** $s_{LND} \simeq 18$ m

Phase	Duration [s]	Power Consumption [W]
Take-off	10	1800
Climb (@ 2.5 m/s)	10	600
Cruise (PL, 5 km)	330	170
Cruise (no PL, 5 km)	390	100
Descent	10	70
Landing	10	70

Table 4: Flight phases simulation.

Stability Analysis After modelling the aircraft and characterising its performance parameters and trim conditions, the team analysed longitudinal static stability, one of the most critical factors in the design of conventional fixed-wing aircraft. The analysis in XFLR5 yielded a stability derivative value of $\frac{dC_m}{d\alpha} \equiv C_{m\alpha} = -0.729$, which is consistent with previous team designs. Subsequently, a dynamic stability analysis was conducted, and the results were found acceptable through comparison with the team’s project database.

Parameters	Short Period	Phugoid	Roll Subsidence	Dutch Roll	Spiral
Frequency [Hz]	1.4748 Hz	0.099 Hz	-	0.60 Hz	-
ξ	0.71	0.013	-	0.199	-
Time to double [s]	-	-	0.03 s	-	5.456 s
τ	-	-	0.04	-	-

Table 5: Dynamic stability analysis.

Control Surface Sizing The sizing of the control surfaces—rudder, elevator, flaps, and ailerons—was optimised to ensure controllability while minimising hinge moments and servo-motor weight. This process involved extracting control derivatives $C_{l\delta}$, $C_{m\delta}$, and $C_{n\delta}$ from the stability analysis, with control parameters set to 1 or -1 based on rotation direction. The Nyx drone served as a benchmark for comparison. Initially, the span was maximised and the chord minimised to enhance control derivatives and reduce hinge moments, with chord increase used as a secondary adjustment. Aileron sizing followed a distinct approach, balancing flap span requirements and considering the hinge line’s alignment with the trailing edge, which caused chord variation along the span. Constraints on servo torque and roll control effectiveness led to iterative refinements, adjusting span and chord until performance criteria were met. The final control derivatives and sizing specifications are detailed in Tables 6 and 7.

Control derivative	Description	Reference value [non-dimensional]
$C_{l\delta}$	Rolling moment control derivative	0.375
$C_{m\delta}$	Pitching moment control derivative	-0.811
$C_{n\delta}$	Yawing moment control derivative	-0.027

Table 6: Control derivatives.

Control Surface	Span	Inner span coord.	Outer span coord.	Chord percentage
Aileron	0.649m	0.868m	1.517m	$\approx 30\%$
Flap	0.753m	0.115m	0.868m	30%
Ruddervator	0.4m	0	0.4m	20%

Table 7: Control surface sizing.

4.2 Aerodynamic Design

The Aerodynamics department focused on optimizing the external shape of O.L.I.V.I.A. to match mission requirements and specifications given by the Structures and Flight Mechanics departments. From the preliminary configuration assessment to the drone polar curves definition, the department aims to improve flight efficiency while assuring satisfying flight quality.

4.2.1 Airfoil Definition

Baseline Airfoil Choice The aerodynamics design starts from the choice of the wing airfoil. A MATLAB script has been implemented in order to screen and filter the UIUC Airfoil Coordinates Database based on targets such as $(L/D)_{max}^{2D} > 80$, $C_{L_{max}} > 1.2$ and $9\% \leq thickness \leq 15\%$. To do so, the code evaluated each airfoil running an XFOIL analysis at an expected Reynolds number (Re) of 300000 for a $0^\circ/12^\circ$ range of angle of attacks (AoAs). The filtered airfoils were then manually analyzed in XFLR5 to select the most suiting one, which resulted to be the S9037 thanks to both its high $C_{L_{max}} \approx 1.4$ and maximum 2D lift-to-drag ratio $(L/D)_{max}^{2D} \approx 93$.

Airfoil Optimisation The S9037 airfoil has been then optimised to better match design requirements with the aid of XoptFoil2 by J. Guenzel (based on the XoptFoil of D. Prosser), setting up a dynamic-weighted Hicks-Henne bump function to operate on eight points at $Re = 340000$. These points were divided evenly among target and optimum ones, both in MTOM and empty expected configurations, to achieve a better lift-to-drag ratio at lower AoAs along with a higher aileron authority (at a local $Re = 150000$ and -2° of expected wing-tip twist). One operational point was set up to optimize the trailing edge section for a flapped configuration, aiming for a higher C_L for 20° flaps at 70% of the chord. The above mentioned points were tuned within a try-and-error process together with the 3D wing definition presented in Sec.4.2.2. The resulting airfoil is the S9037_{UAS25} presented in Fig.2, which shows a relatively significant improvement in 2D lift-to-drag ratio (+10 points circa for most flying AoAs) and C_L values (+25%).

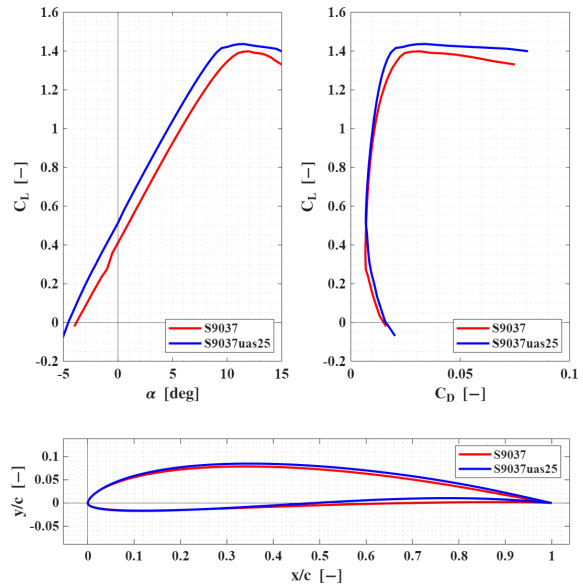


Figure 2: Optimized versus baseline airfoil.

4.2.2 3D Wing Definition

Geometry Definition and Optimization Starting from an Aspect Ratio (AR) of 12 and a wing surface (S) of $0.85m^2$, a baseline rectangular geometry was taken as a reference in the XFLR5 analysis for further improvements. The main aim was to achieve a theoretically elliptic local lift distribution by testing several combinations of taper ratios, wing sweeps and wing twists, all while keeping AR and S constant. The overall shape underwent back-and-forth adjustments between the Aerodynamics and Flight Mechanics departments to also assure satisfying flight performances, in particular related to wing stall and flight dynamics modes recovery. The chosen configuration consisted in a semi-elliptical wing with a straight trailing edge, which assured an almost-elliptical lift distribution along with more ease of manufacturing for the control surfaces, and a mid-to-tip twist of -2° to improve aileron authority and recover loss of lift distribution efficiency while avoiding tip stall.

CFD Validation The XFLR5-defined wing was then reconstructed on a CAD model and coupled with the first iteration of the fuselage shape. The said geometry was then studied through a preliminary CFD analysis using Ansys Fluent (2024 R2 version), to validate and correct the panel-method-based results and to evaluate the fuselage-wing interaction, along with the flow detachment behavior on the lower-aft side of the body. The results were coherent with what was seen on XFLR5, taking into account the high amount of drag introduced by the fuselage bluff body.

4.2.3 Future Improvements

Wingtips Together with the Structures department, it was considered that the narrow wingtip section was too small to be manufactured with carbon fiber laminate, and thus a 3D-printed tip was reputed to be a better option. This paved the way to wingtip shape optimization due to the shaping freedom and fast prototyping advantages intrinsic of 3D-printing. Up to this project phase, only a Hoerner-tip kind of wingtip has been studied through CFD analysis, but it has not been deemed satisfying in terms of drag reductions; thus, more studies in this matter are planned, with the aim to test with the aid of CFD various wingtip and winglet designs, in order to further improve the wing efficiency.

Landing Gear Fairings A tricycle landing gear configuration has among its disadvantages the one of having a third bigger gear leg which contributes to increase parasite drag. In the attempt of mitigating this issue, landing gear fairing options were studied once again through CFD analysis. Two main configurations were tested and compared with the naked wheels reference: one based on literature (NACA report No.485 by W.H. Herrnstain and D. Biermann, 1934), and one consisting on a revolved-NACA0030 airfoil, both taking into account the ground clearance for the take-off expected angle on a grass runway. The results have been satisfying for both these two configurations, with a C_D reduction compared with the naked wheels of -15.67% and -20.46% respectively. As of now, a trade-off study on whether the efficiency improvement is worth the added weight is being sustained between the Aerodynamics and Structures departments, and a decision is expected to be taken during the flight testing phase.

4.2.4 Polar Curves

Configurations For the final definition of aerodynamic and flight dynamic properties of O.L.I.V.I.A., several CFD simulations have been used. To have an utter idea of the drone performances in all flight conditions, three main configurations were chosen: take-off (25° flaps), cruise (*clean*), landing (30° flaps). The aimed speeds for these three are, respectively, 12, 18 and 10m/s. The range of angle of attacks considered have been $0^\circ/12^\circ$ for take-off, $-4^\circ/12^\circ$ for cruise and $0^\circ/10^\circ$ for landing.

Setup of CFD Analysis A grid convergence study was prior conducted to ensure a good reliability of the results. It has been developed by assuming fixed (half-)domain dimensions while varying mesh growth parameters in order to increase the number of cells (and nodes); a range of 0.7/8.5 millions (M) cells meshes have been considered and appropriate refinement boxes were placed in order to increase cells allocation efficiency. The target results (C_L and C_D) came to convergence at 5M mesh cells circa, which seemed a good trade-off between results accuracy and relatively limited computational resources. All the steady RANS simulations were performed on Ansys Fluent 2024 R2, using a SST k- ω turbulence model and second-order upwind schemes for the pressure-based solver; to bring down computational costs, only half of the domain has been taken into account, thanks to the symmetry of the conditions that have been simulated.

Results The CFD analysis results were found to be consistent with the expected flight characteristics of O.L.I.V.I.A.: the maximum lift-to-drag ratio is achieved at $\alpha \approx 6^\circ$; the values of C_L during both take-off and landing satisfy the competition requirements; and $C_{L_{max}}$ is sufficiently high to allow for a wide flight envelope. In conclusion, the aerodynamic design is considered to be well-performing and does not require any adjustments. Further analysis will be conducted to investigate the potential benefits of landing gear fairings and wingtips in enhancing O.L.I.V.I.A.'s performance.

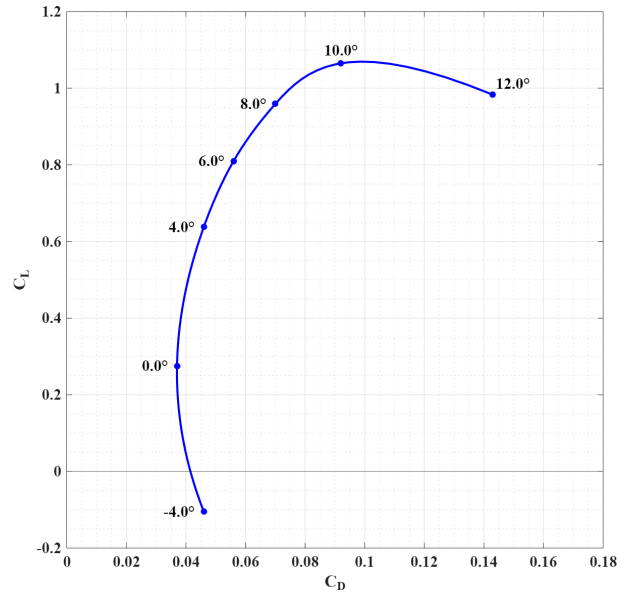


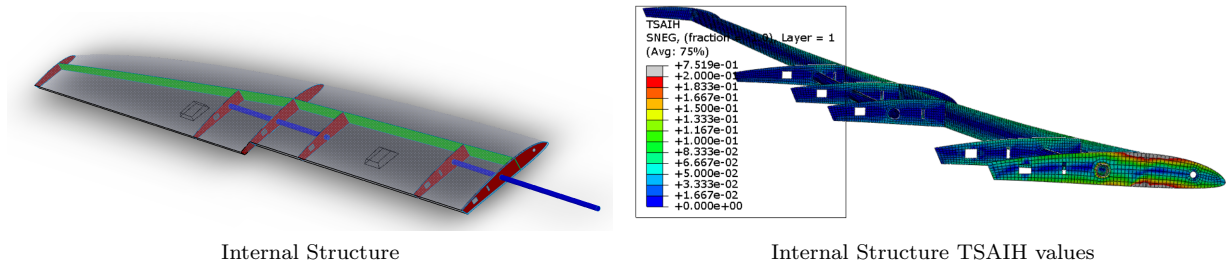
Figure 3: Clean configuration polar curve.

4.3 Structural Design

Structures department is responsible for the design of O.L.I.V.I.A., ensuring compliance with regulations, necessities, and results provided by the other departments while minimising weight and maximising structural performance; all while keeping an eye on the feasibility of its construction. Therefore, tasks range from choosing manufacturing processes, FEM simulations, and materials selection, to studying and comparing lamination sequences, developing kinematics for payload release system, and overseeing manufacturing and assembly processes.

4.3.1 O.L.I.V.I.A.'s Structure

Wings



Wings are composed of five sections, of which the central one integrated in the fuselage. Each wing is connected to the fuselage via carbon tube and a pin, which constrains the rotational and translational degrees of freedom in all directions. The assembly is secured with two drawbolts, one per side, positioned on the lower surface of the airfoil to prevent extraction.

The connection tube runs through the fuselage, to evenly distribute loads. The inner (root) and outer (tip) sections of each wing are joined through bonded interface ribs, with an additional tube and pin system enhancing load distribution for optimal structural performance. Wingtips are placed at the tips' end, with one housing the instrumentation for the pitot system.

In order to maintain the airfoil's geometry while providing structural support to the connection tube, the internal structure consists of 16 ribs, three at wing-fuselage interface and two at each other interface, spaced 16 cm apart. A spar runs throughout the span, ensuring better response to bending loads.

Ribs, electronic components boards and fuselage bulkheads are cut via waterjet from a carbon fabric and *Nomex* honeycomb core sandwich plate. The spar is also a sandwich structure, composed of both carbon fabric and unidirectional layers, with a *Rohacell* foam core. Foam (isotropic) and honeycomb (anisotropic) have been used in a way that maximises $\frac{\Delta \text{Moment of Inertia}}{\text{Weight}}$.

Each wing section is constructed using carbon fibre monocoque design, where the skin acts as the primary load-bearing structural element. This approach minimises and lightens internal framework, whose structural contribution is primarily supplementary.

On the lower surface of each section of the wings, a recessed compartment houses the servomotors, which controls the movable surfaces hinged to the wing.

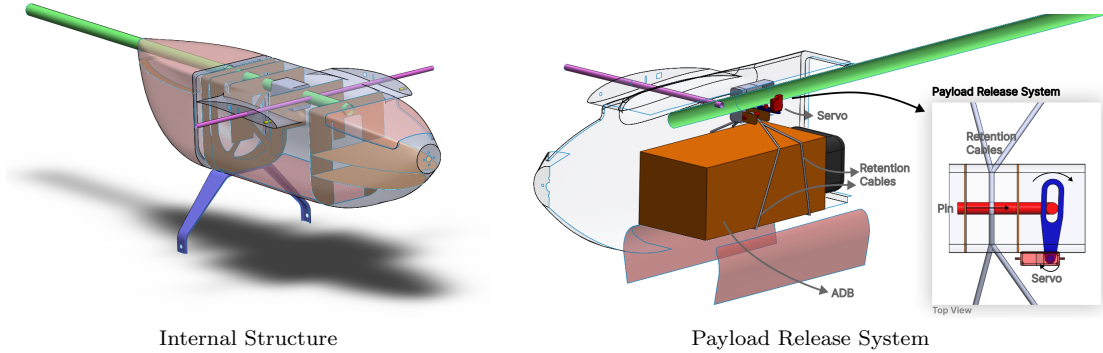
Fuselage

The fuselage structure is divided in two main sections: the front section, made of carbon fibre and designed to accommodate the payload and electronic components, and the rear capote, made of glass fibre to meet operational requirements for GPS placement. The front fuselage structure is divided into two parts: the upper and lower sections. This allows for greater convenience in laminating the complex internal framework. Unlike wing structure, in fact, fuselage internal framework is critical for overall structural behaviour due to the presence of various large openings, such as payload release bay and front and rear capotes, that greatly reduce the monocoque structural effectiveness. Nonetheless, the skin is composed of multiple sections with different material configurations, ensuring increased strength in the highly stressed areas.

The internal structure consists of five honeycomb and carbon fabric sandwich bulkheads, positioned transversely to the symmetry plane of the fuselage. The first and fourth reinforce payload bay's perimeter, while two intermediate ones prevent horizontal movement. A final bulkhead, positioned just behind the fourth, strengthens the fuselage underside at the landing gear attachment point. The tail tube, which

connects the tail to the fuselage, passes through the bulkheads and is secured to the fuselage via the wing tube that runs through it.

On top of the tail tube, attached to the top surface of the fuselage, between the 2nd and 3rd bulkhead, a carbon fabric and *Rohacell* case with reinforcement ribs is placed to house Payload Release System kinematics and servo motor. As described in Concept Paper, a wire wraps both the payload and a pin, that when retracted frees the wire and allows the payload to drop. In the front section, near the front capote, carbon fabric and honeycomb sandwich platforms support the electronic components.

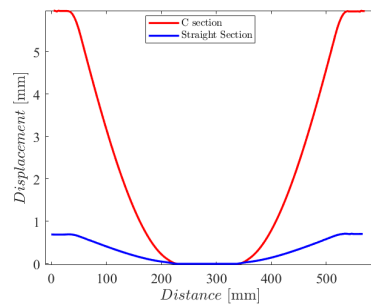


Internal Structure

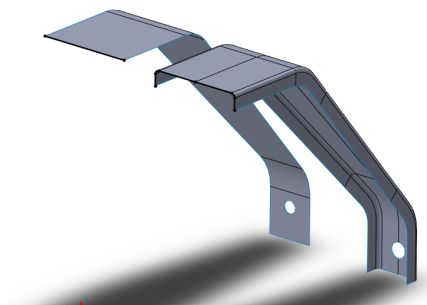
Payload Release System

Landing Gear

The rear landing gear has a C-shaped cross-section. Structural studies on landing gear geometry have shown that, with the same setup sequence, the C-shaped section exhibits considerably lower deformation compared to a conventional straight section, which is critical to ensure propeller clearance from the ground and stable take-off and landing due to increased damping. The lamination plybook includes unidirectional and carbon fabric plies orientated at 45° . The nose wheel will be made of aluminium and will feature a steering mechanism with a corresponding servomotor.

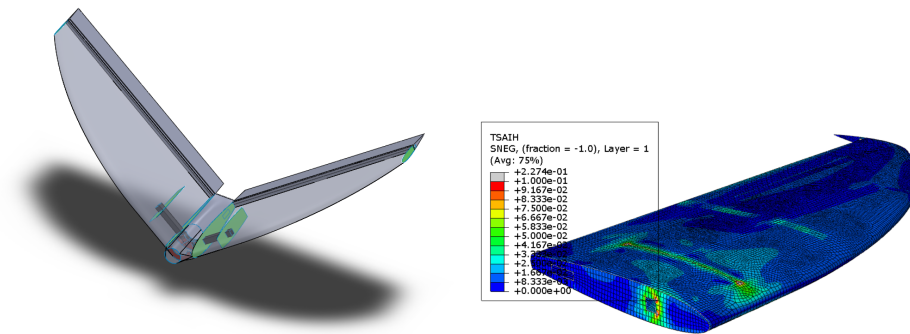


Straight vs C Section Deformation



Sections

Tail



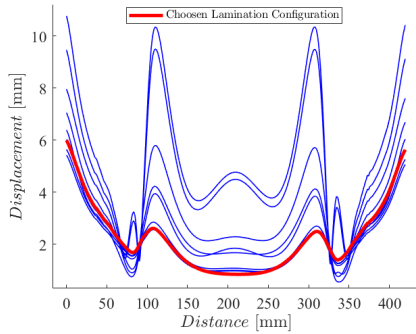
Internal Structure

Sections

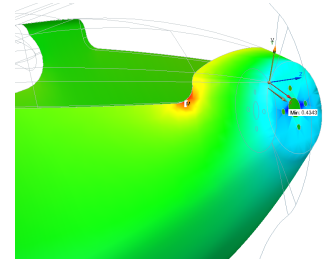
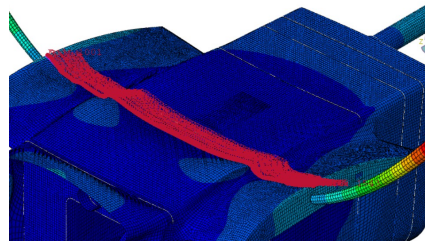
Tail is designed as a V-shaped tail, composed by two drifts and a hub that connects them to the tail tube. Drifts' carbon fibre monocoque structure is similar to the wing's one. Internal structure consists of three sandwich ribs: one at the interface with the hub, one as support to the connection tube, and one to close the wingtip. Due to the relatively low loads acting on the tail, spar is replaced with spanwise unidirectional carbon fibre reinforced areas.

Lamination Sequence and Internal Structure Optimisation

In order to achieve the best trade-off between structural performance and minimal weight, several FEM simulations and calculations were carried out. Composite materials were chosen for their mechanical properties and lightness (for detailed description see Manufacturing section), along with lightweight core materials, to increase the thickness and moment of inertia of the thinnest parts. The figures below show an example of simulation results comparison of the upper fuselage lamination sequence. FEM results were compared by analysing the TSAI-Hill and TSAI-Wu values according to the specific design requirements, such as the minimum thickness for bonding and plies drapability.



Displacements comparison of different upper fuselage lamination sequences



Plies wrapability study

Weight Breakdown

The weight estimation was periodically updated using Excel, segmenting the structure into major components and subcomponents while recording their surface areas based on CAD models. These values were multiplied by the density of the composite fibre, resin, or other materials, with an additional error margin accounting for assembly adjustments, bonding, and unforeseen factors, varying by component type and based on previous years' experience. This approach allowed for automatic updates to the aircraft's weight estimation whenever variables changed. The final weight values, with and without the error margin, were then combined with the estimated electronics weight provided by the EPC department. Below is an example of the Excel matrix used, along with a summary table presenting the calculated weights in a more compact format.

				Fabric	Fabric	Fabric	UD	UD	Rohacell	Glass	Honeycomb					
			Nominal Density	62	162	165	100	150			100					
			Resin %	55	42	40	40	40		50						
Uncertainty + Other	Part	Sub-Part	Surface (m ²)	112,7273	385,7143	412,5	250	375	110	200	100	Wgt (g)	Wgt (g) + Error	Total (g)	Payload (g)	Structure Wgt%
0,25	Fuselage	Bulkheads	0,1330213			2										
		Upper Surface	0,12526547		4											
		Lower Surface	0,06553991	8	2											
		Ribs	0,01124169	8	2					5						
		Stiffening Frames	0,02820015		1			2		1						
		Motor Mount Front	0,0036191	2	1					5						
		Motor Mount Side	0,00959312	2	1											
		Front Capote Stiffener	0,02233531		2			1	2							
		Main Skin	0,34209925		1											
				0	109,7426	0	0	0	0		13,30213	676,6258	845,7823			1141,537096

Component	Weight [g]	Uncertainty Percentage
Fuselage	845.78	25%
Root Wing	537.65	10%
Tip Wing	373.23	10%
Tail	114.04	10%
Flaps	106.81	0%
Ailerons	49.89	0%
Capotes	187.22	20%
Wing Tubes	299.91	0%
Tail Tube	392.79	0%
Electronics	1300	0%
Landing Gear	395.67	50%
Total	4455.15	
Payload	2544.85	

Table 8: Weight breakdown.

4.4 Electronics Propulsion and Control

The EPC (Electronics, Propulsion & Control) department is responsible for the design and selection of COTS (Commercial Off-The-Shelf) components for all electronic systems and the propulsion power unit on board O.L.I.V.I.A. This year, the main focus was on the development and testing of the novel autonomous drone system, which is powered by the **ArduPilot** firmware. COTS components have been carefully selected to ensure the highest level of compatibility with the mission, as well as to provide the safest and most redundant solutions while optimising weight and power consumption of these systems. The department has also conducted extensive testing and studies to select the optimal combination of propulsion components (battery, ESC, motor, and propeller). This combination aims to make the plane as lightweight as possible while ensuring better efficiency and complying with the competition's constraints.

4.4.1 Electronic Components Selection

At the beginning of the project, efforts were primarily focused on selecting the most logical and cost-effective solutions to ensure the success of the O.L.I.V.I.A project. For each COTS component, many options were available on the market. The table below presents each component selected for the on-board systems, along with the main reasons behind each choice.

Component	Selected COTS	Reasons behind the choice
Electric Motor	T-Motor AT4120 KV560	Higher power-to-weight ratio, high efficiency, compatible with required performances at take-off.
Electronic Speed Controller (ESC)	T-Motor C80A Single ESC	Compatible with high currents required by motor at take-off, reliable heat dissipation, lightweight, telemetry included.
Battery	Gaoneng GNB LiHV 6S 3000mAh 120C	Highest energy density among the competitors, high C rate, low internal resistance, suitable capacity for the mission (tested at bench).
Flight Controller (FC)	Mateksys H743-WING V3	Best balance of weight, cost, sensor redundancy, number and versatility of serial ports; best microprocessor on the market (STM32H743); powerful built-in BEC; widely supported; compatible with ArduPilot.
GNSS and Compass Module (x2)	Mateksys M10Q-5883	Best performances in number of satellites and less time to lock them, precise compass, lightweight, good price and reliability.
LiDAR Rangefinder	Benewake TFmini Plus	Best balance between precision, cost, reliability; lightweight; IP65 rating; sufficient range for smooth landing.
Airspeed Sensor	Mateksys ASPD-4525	High precision and reliability, best value for money, lightweight, perfect compatibility with FC.
Servomotors	MKS HV6120 (wing) MKS HV70 (tail) MKS HV6150 (front wheel)	Sponsored, very high quality, high precision, support various operative voltages, and, above all, one of the highest torque-to-weight ratio on the market.
Radio receiver	Radiomaster RP4TD-M ExpressLRS 2.4GHz Mini True Diversity Receiver	Best performances in terms of range, signal-to-noise ratio, reliability and stability of the radio link; lightweight; dual channel true diversity; good bandwidth; ExpressLRS protocol.
Ground station receiver	Mateksys mR24-30 2.4GHz Receiver	Not compliant with CE, replacement for this component is still being identified.

Table 9: Summary of selected components.

4.4.2 System Architecture and Data Flow

Figure 4 presents the system architecture and data flow for navigation, mission control, flight control, and sensor integration, enabling autonomous operation. The flight controller processes mission plans, waypoints, and geofencing constraints while managing actuators—servos and electronic speed controllers (ESC)—based on sensor data from the Pitot tube, IMU, LIDAR, and GPS. Pilot input determines the flight mode, switching between normal and autonomous flight. A feedback loop continuously updates the flight controller for real-time stability and navigation adjustments. A failsafe mechanism ensures a return to normal flight in case of system anomalies, maintaining a balance between autonomy and safety.

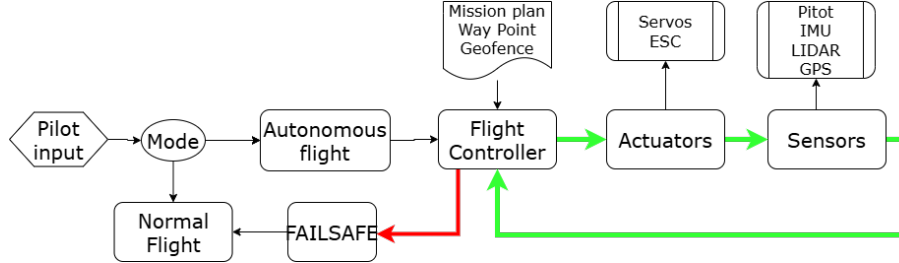


Figure 4: System architecture and data flow.

4.4.3 Propeller Selection

The propeller selection for O.L.I.V.I.A.’s competition flight remained unresolved until this stage. The key factors influencing this decision are the take-off distance, favouring a low-pitch, large-diameter propeller, and the maximum propulsive efficiency during cruise, optimised with a high-pitch, small-diameter propeller. Therefore, the selection required a trade-off between these conflicting factors. A two-phase approach was adopted:

1. Development of a **MATLAB** algorithm to model motor and propeller performance.
2. Static thrust testing campaign to validate the algorithm, and select the most efficient propeller (in simulated cruise conditions) that provides the necessary thrust during take-off.

The algorithm was initially developed due to unforeseen issues with the university’s wind tunnel, which became inoperable just before scheduled tests. This led to the creation of an efficient alternative, which has since proven to be a valuable tool also for future projects. The algorithm identifies the optimal APC propeller for a given motor, ensuring that thrust requirements for take-off are met while minimising electrical power consumption during cruise. The selection process follows these steps:

- The script retrieves all available propellers from the APC database.
- Compatible motor-propeller coupling is identified by checking if the corresponding power curves intersect each other with acceptable thresholds. If no intersection occurs, the propeller is discarded.
- Static thrust at maximum throttle is computed, and propellers that do not meet the required take-off thrust (60 N) are excluded.
- For the remaining propellers, efficiency in achieving the required thrust at a given airspeed is calculated. The corresponding throttle (PWM) is determined for cruise conditions.
- Motor operating conditions are used to compute the current, voltage, and electrical power required.
- Propellers are ranked based on their power efficiency during cruise (≈ 14 m/s).

The algorithm identified the **APC 15x8E**, **15x10E**, and **16x8E** propellers as optimal, consistent with the expected results for the *T-Motor AT4120*.

Despite the algorithm’s high reliability, uncertainties remain due to potential APC data inaccuracies and motor model limitations. To validate the results, static tests were conducted using the department’s test bench, equipped with load cells and interfaced with an **Arduino Nano** via a custom PCB. The process involved motor model calibration at various throttle inputs, refining torque curves by comparing measured and theoretical values. Full-power tests then verified the take-off thrust of optimal propellers.

The propellers were further tested under maximum power conditions (80 A) on the test bench and using the algorithm. The results are presented in Table 5 and Figure 6, illustrating that the *APC 15x8E* is the optimal choice, providing sufficient thrust during take-off while exhibiting the highest propulsive efficiency at cruising speeds.

Propeller	T_{TO} [N]	η_{cruise}
15x8E	69	0.718
15x10E	60	0.686
16x8E	74	0.633

Figure 5: Propeller performance data.

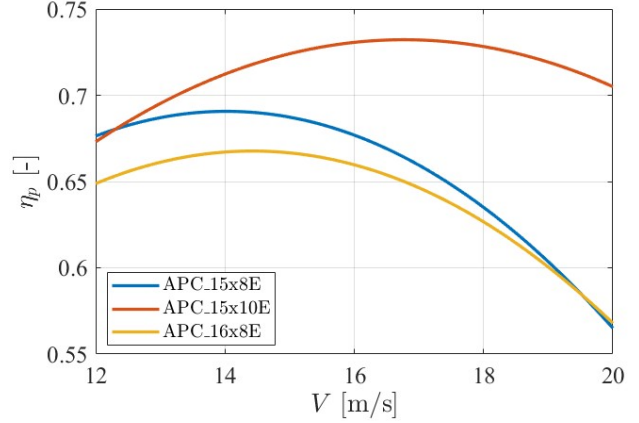


Figure 6: Propeller efficiency over airspeed.

4.4.4 ESC Cooling System Evaluation

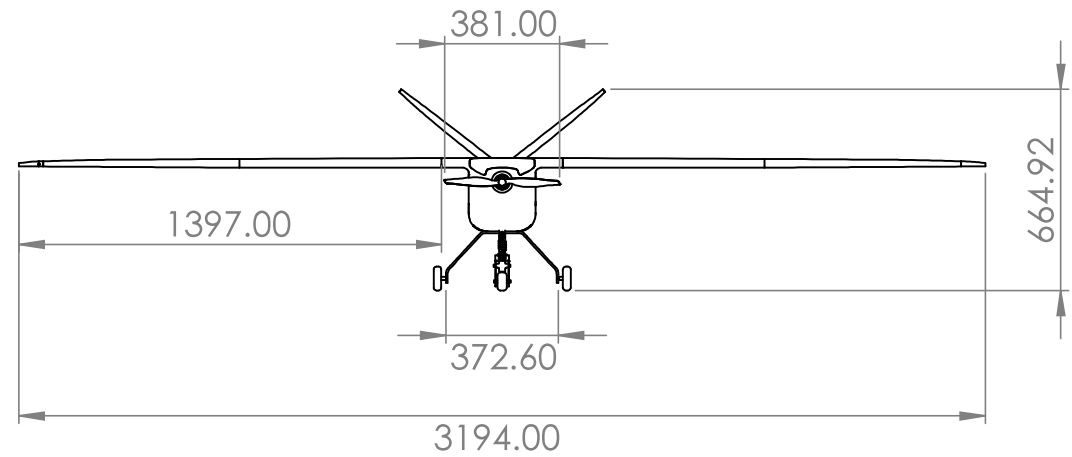
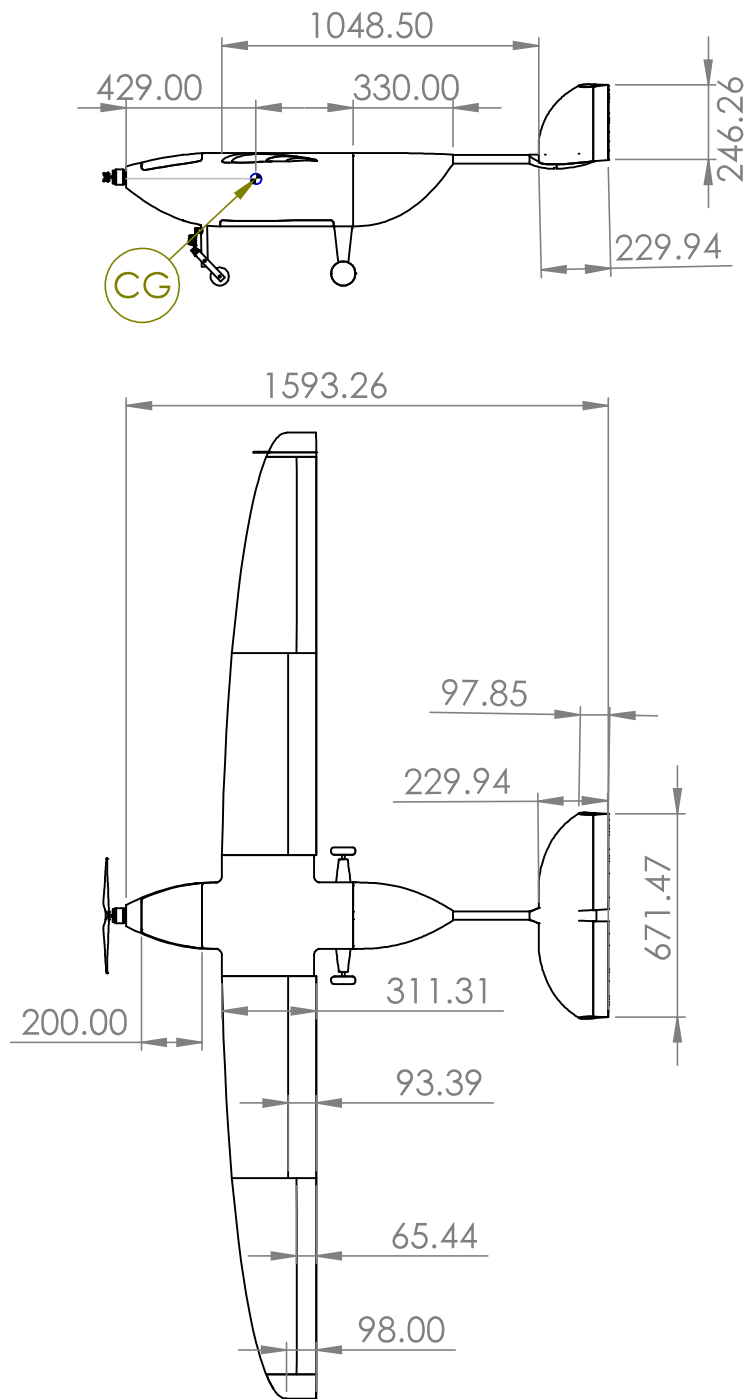
The necessity of an active cooling system for the electronic speed controller (ESC) was assessed due to the high-power requirements of O.L.I.V.I.A.'s propulsion system. As the current draw is significant, thermal stress on the ESC must be managed while maintaining minimal weight. To evaluate whether passive cooling is sufficient, a series of laboratory tests were conducted under simulated flight conditions, including cruise and take-off. Worst-case conditions were simulated by enclosing the ESC in a sealed compartment with limited heat dissipation. Temperature data were recorded using ESC telemetry and validated with an infrared (IR) sensor. Cruise power consumption of approximately 200 W was estimated using aerodynamic data and component efficiencies ($\eta_{\text{propeller}} = 0.7$, $\eta_{\text{motor}} = 0.85$, $\eta_{\text{ESC}} = 0.9$). 15-minute tests were performed with a power supply, confirming that the ESC remained within acceptable thermal limits without active cooling. Take-off requires a thrust of 60 N, corresponding to an estimated 80 A current draw. Given the short duration of this phase ($\simeq 15$ s), ESC temperatures were monitored during full-power testing, demonstrating a peak below 82°C. Results indicate that passive cooling is sufficient, as temperatures remained well below the ESC's maximum limit of 110°C. Despite higher transient temperatures during take-off, the short duration prevented excessive heat accumulation, ensuring safe operation without active cooling mechanisms.

4.4.5 Battery Testing

The battery testing aimed to verify whether the batteries selected during the preliminary design phase, based on estimated mission power consumption, were suitable for the competition scenario while ensuring at least 20% charge remained upon landing, as required by the regulations. The primary goal was to confirm that the stored energy was sufficient to complete the mission while maintaining optimal performance. The tested battery candidates were the *Yowoo LiPo 3300 mAh* and the *GNB High-Voltage LiPo 3000 mAh*. To ensure consistency, both were discharged on a static test stand using the competition motor and the same propeller. The mission scenario was simulated based on Table 4, with assumptions including $\eta_{\text{propeller}} = 0.7$, $\eta_{\text{motor}} = 0.85$, and $\eta_{\text{ESC}} = 0.9$, ensuring realistic power consumption. Data exported from the flight controller and processed in MATLAB included key battery metrics such as voltage, current discharge, energy output, and internal resistance. The test did not account for servo power consumption or a potential go-around, leading to a conservative interpretation of results. Both batteries were deemed suitable for safe flight, with the final choice favouring the *GNB High-Voltage LiPo 3000 mAh* due to its superior weight-to-energy efficiency ratio.

Battery	Weight [g]	Initial Voltage [V]	Final Voltage [V]	Charge [%]
<i>Yowoo Lipo 3300 mAh</i>	533	24.6	23.0	45
<i>GNB HV Lipo 3000 mAh</i>	403	26.1	23.1	46

Table 10: Battery performance.



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all measurements are in mm

5 Safety

This section aims to summarise and explain the overall approach to safety and the establishment of the airworthiness of the UAS. An engineering holistic approach was followed, systematically analysing the mission type, operational context, and human and time-related factors, in alignment with local and European guidelines as well as the CAA Guidance, CAP-722, as suggested in annex C.9 of the Regulations. Given the non-commercial nature of the project and the limited expertise of the student team, a deterministic risk assessment was deemed unreliable, as it would likely fail to account for all possible scenarios. Instead, a probabilistic quantitative risk assessment was conducted using risk matrices and a retrospective analysis of past incidents and contingencies.

To maximize clarity and information density, data will be presented in tables and graphs. For further details on the referenced tests and analysis, see the Design Description and Qualification Test Plan sections.

5.1 Airworthiness

To ensure the airworthiness of the aircraft, specific considerations and design choices were made during the development phase. These were subsequently tested both analytically using software and through practical tests. These considerations aim to guarantee the reliability of the structure, control and propulsion system, and emergency procedures, while also facilitating and making as safe as possible all operations related to the construction, assembly, and maintenance of the aircraft. These considerations are summarised in the table below.

Once the design was assessed and the associated risks identified, the safety analysis discussed in the following sections was carried out. A fatigue life and maintenance evaluation will be conducted once the structure is completed and the aircraft is assembled, in order to use O.L.I.V.I.A. for further upgrades and in future competitions. Once the UAS reaches the end of its fatigue life, it will be used for educational purposes within the association and in the classroom.

Airworthiness Evaluation				
Unit	Design Phase	Test Phase	Results	Pending Actions
STR & Materials	FEM analysis, strategic breaking points identification, assembly Study, industrial partner's confrontation, adequate safety margin in plybook.	Stress test on similar structures, failure test of carbon tubes and 3D printed parts.	Plybook that ensures structural integrity under worst flight conditions (safety margin of 1.5), simplest possible assembly procedure definition, carbon tubes and 3D-printed parts capable of withstanding flight loads definition.	Lamination quality check, lamination tests on carbon tubes, non-destructive testing on assembled structure, creation of spare parts (greater flexibility in flight testing).
Control & Communication	Two separate links: one dedicated hardware for the radio and one for the ground station, RP4TD-M ExpressLRS 2.4GHz Mini True Diversity selection, ExpressLRS protocol, failsafe configuration, pre-flight checklist definition.	Testing of all electronic components, thrust stand tests of the propulsion system (ESC, batteries, motor, propeller, BEC) beyond normal operating conditions, PID controllers tuning on flight axes, turn and navigation parameters adjusting, flight sensors calibration, autonomous waypoints mission testing, hand-launch take-off testing.	Two radio links ensure redundancy and communication reliability, True Diversity system enhances range and link stability, ExpressLRS protocol provides superior latency and range performance, failsafe system in ArduPilot successfully configured, all electronic components function correctly, propulsion system performs reliably under high load conditions remaining within operational limits, ESC within safe temperature limits.	Autonomous take-off and landing testing, payload release testing (initially simulated with X-Plane + Mission Planner and later during flight), TECS tuning (speed and altitude control improvement), non-destructive test of failsafe.

5.2 Risk Assessment and Bow Tie Diagram

The risk assessment was conducted alongside the airworthiness evaluation, using risk matrices with severity and frequency definitions from Annex D.3 of the regulation. The acceptability level follows the Roland and Moriarty (1990) classification, adapted to these definitions. Due to space constraints, only the most critical risks are shown. They are represented as dots placed within the matrix based on their classification *before* safety measures. Arrows indicate their classification *after* the implemented safety measures, summarized in the table below. To further clarify the adopted safety process, a bow-tie diagram (adapted to this context) illustrates the safety barriers effects in a Major Crash event.

Hazard Classification (adapted Roland and Moriarty, 1990)				
HRI	I - Marginal	II – Minor	III – Major	IV - Catastrophic
A - Frequent	1 A	2 A	3 A	4 A
B - Occasional	1 B	2 B	3 B	4 B
C - Remote	1 C	2 C	3 C	4 C
D - Improbable	1 D	2 D	3 D	4 D

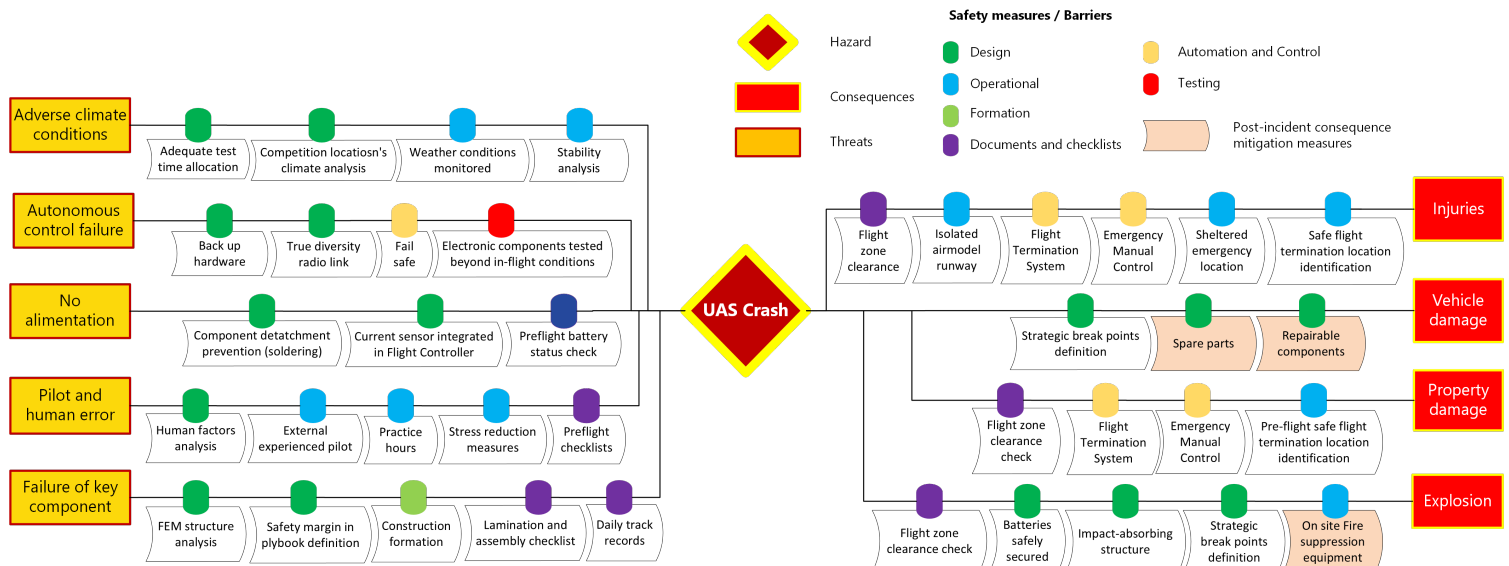
Frequency Definition	
A - Frequent	Likely to occur frequently during UAS Challenge.
B - Occasional	May occur occasionally during UAS Challenge.
C - Remote	Remote possibility of occurring during UAS Challenge.
D - Improbable	Highly unlikely to occur during UAS Challenge.

Severity Definition	
I - Marginal	Irreparable damage or loss of the UAS.
II – Minor	Minor injury to a participant. Damage to public property.
III – Major	Single major injury to a participant. Single injury to a member of the public.
IV - Catastrophic	Multiple injuries. Death of any party.

Acceptable without project revision	Undesirable	In flight collision	Frequency interference / Control system hacking	Premature payload release	Component failure
Acceptable after project revision	Unacceptable	Major crash	Communication issues / Signal loss	Loss of control / Motor failure	

Severity and Frequency Mitigation Measures

Measure	Effect
Strategic breaking points definition	Structural failure control, aircraft damage minimization, repair planning.
Impact absorbing structure/battery fastening	Battery shocks prevention, risks reduction of cuts, explosions, damage and injuries.
Spare parts and repairable components	Reduction of aircraft loss risk during flight or assembly.
Safe zone identification for FTS	Minimising damage and reducing emergency pilot stress.
Flight zone clearance and preflight checklists	Impacts on unintended people or objects prevention.
Geo-Fence recognition testing	Ensures FTS activation in designated flight zone, avoiding injuries and damage.
Electronic components testing	Lower failure risk, ensures the emergency systems' proper functioning.
Back-up hardware	Receiver failure control: dual-channel data distribution boosts safety and bandwidth.
True diversity Radio Link	Improves range, reliability and control, mitigating risk frequency and severity.
Weather condition monitoring	Prevents failures worsening and loss of stability and control.
Current sensor	Excessive draw detection and battery discharge prevention.
Flight Termination System	Prevents uncontrolled flight from causing harm outside the flight zone.
Emergency manual control	Greater aircraft control in autonomous failure, mitigating risk to people and objects.
Stability analysis	Prevents crashes caused by aircraft instability.
Frequencies certification from manufacturer	Reduces frequency interference risk.
Frequency interference legislative provisions	Lowens the likelihood of frequency disruption and control system hacking.
Cable detachment prevention (soldering)	Reduces the likelihood of control system failure due to cable disconnection.
ExpressRLS protocol	Superior performance in latency and range, reducing loss-of-control probability.
Servo Oversizing	Prevents overheating and servo failure, avoiding aircraft loss of control.
Payload Release System (PRS) testing	Reduces premature payload release risk.
FEM analysis & safety margin in plybook	Decreases structural failure risk of the payload support and structure.
Propulsion system extreme-condition testing	Lowens the risk of motor issues or instability due to induced torques.
Construction checklists & formation	Reduces in-flight failures from lamination or assembly errors.



6 Manufacturing and Support Description

6.1 Manufacturing

6.1.1 Composite Lamination

Due to their lightweight properties and versatility, most O.L.I.V.I.A. components are manufactured using composite materials. These materials offer an excellent balance between design freedom and technical performance, and the team's experience with composites ensures proficiency in lamination. To maximise lamination efficiency, the shifts were arranged to ensure continuity of workflow. Before shifts began, a training workshop on lamination techniques was conducted and comprehensive reference documents were provided for easy access at any time. In addition, checklists were introduced to support the workflow, minimising the risk of errors and consequently reducing time and material waste. Composites components are fabricated using various techniques and materials, including woven prepreg carbon fibre (CF), unidirectional prepreg carbon fibre (UD), dry Kevlar, Rohacell, and honeycomb structures. Below is a brief description of the lamination process for each component.

Wing Lamination Wings were produced using ten moulds: two for control surfaces and eight for the four wing sections, assembled in pairs (upper and lower surfaces) and cured closed. This complex process involved layering a base of CF, placing a spar made of UD and Rohacell on a double UD reinforcement, and securing it with an additional ply of CF. Rohacell reinforcements wrapped in UD were positioned along the span and at the wing attachment points. The lower ply was extended to create an overlap that, once the moulds were closed, bonded to the inner surface of the upper section, preventing gaps. Two tubular internal vacuum bags run through the closed mould on either side of the spar and are connected externally to the main vacuum bag for autoclave curing. For the hinges, a strip of dry woven Kevlar was placed along the trailing edge and pressed with Kraiburg rubber, ensuring flexibility even after the autoclave curing process, which helps maintaining smooth airflow over the wing. For the support tools mentioned in Chapter 4.3.1, a CF and *Nomex* honeycomb sandwich plate has been laminated.

Tail Lamination The tail lamination involved 2 moulds for tail empennages, which were cured in a closed, and a separate mould for the connector, also laminated closed. Tail's plybook includes two base plies of CF, two longitudinal UD reinforcements, and one transverse reinforcement in CF and Rohacell.

Fuselage Lamination The central fuselage will be produced using two moulds (for left and right halves) which will be assembled and cured closed. The fuselage plybook will include multiple CF plies, with Rohacell and UD sandwich reinforcements. An additional mould will be used for the upper section, while two more will be employed for the rear cover, also cured closed and laminated with glass fibre, given its purely aerodynamic function.

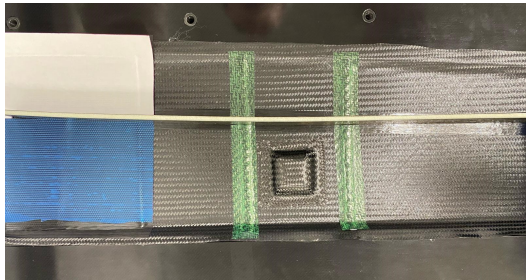


Figure 7: Wing lamination sequence.



Figure 8: Internal wing vacuum bags.

6.1.2 Additive Manufacturing

3D printing allows for rapid prototyping, enabling the team to experiment autonomously with different shapes and materials. This flexibility facilitates testing various filaments and optimizing designs, while also simplifying the production of spare parts when needed. Three main components were manufactured using 3D printing: the tail fairings, the servo mounts, and the drawbolts. Two printing techniques were employed based on the mechanical requirements of each part. The tail fairings were produced using vase mode, a technique in which the printer forms a continuous spiral structure in a single pass. This method, combined with lightweight PLA filament, ensures minimal weight. The fairings were reinforced with a carbon fibre rod, which also serves as a hinge for the control surfaces. For servo mounts and drawbolts, traditional infill printing was used to provide the necessary mechanical strength and rigidity. This approach ensures durability, particularly for load-bearing components such as the control arms of the tail surfaces and the servo motor supports.

6.2 Final assembly

After lamination, all components are trimmed to their final dimensions. Then, the bonding surfaces are sanded with sandpaper and cleaned with acetone.

Bonding is done with structural grade epoxy resin, chosen for its high strength, suitable curing time, and compatibility with carbon fibre composites. Ensuring a safe working environment is a priority throughout the assembly process. The workspace is well ventilated and appropriate personal protective equipment (PPE) is used: particulate filtering masks, cut-resistant gloves and goggles when cutting carbon fibre and vapour-resistant face masks when handling chemical agents.

Wing Assembly Firstly, servo motors are installed and electronics wired. Then, the carbon fibre sandwich ribs are bonded in place. Since wings and fuselage are manufactured separately and later connected via a carbon tube, precise alignment of the ribs is essential. To address this challenge, custom 3D-printed jigs have been developed to ensure correct positioning. Control surfaces are then attached to the Kevlar hinges. Once the control surfaces are in place, the control rods are installed and actuators are tested. Finally, each wing is mounted. The outboard and inboard sections of each half of the wing are joined by inserting a carbon fibre rod through the ribs and bonding the interfaces together.

Landing Gear Assembly The nose landing gear is assembled by attaching the strut to the swing arm, installing the spring shock absorber, and finally mounting the wheel onto the axle. The main landing gear is constructed by bonding the axles to the main structure before installing the wheels.

Fuselage Assembly The fuselage consists of two primary sections: the main body, which houses the payload and electronics, and the wing section, which transfers aerodynamic loads from the wing to the main structure. Assembly begins by installing the bulkheads within the main body, securing the payload, electronics, and tail tube attachment points. Special care is taken to ensure the precise alignment of the tail. In the wing section, additional ribs are inserted to support the spar tube, which connects the wings. Once these elements are in place, the wing section is bonded to the main body, forming a single integrated structure. Following this, motor, electronics, and payload release system are wired to the main body. Two drawbolts are then bonded to the underside of the wings and the wing section of the fuselage to secure the wings. The nose landing gear is then secured using Seeger clips, allowing it to rotate for steering, while the main landing gear is bolted to the underside of the fuselage.

Tail Assembly Tail assembly consists of the empennages and the hub, which connects to the tail tube. The empennages are reinforced with internal ribs and a square tube that acts as both the structural spar and the connection to the hub. In the hub, internal structure is bonded, and servo motors are wired. Once the main components are assembled, the 3D-printed rudder fairing is attached to the empennage using superglue, along with the fairing for the hub. The hub is then connected to the tail tube using two screws that secure into 3D-printed nuts inside the tube. Finally, the rudders and empennages are installed by inserting a screw into a nut positioned at the end of the spar tube in the empennage.

6.3 Storage container

The storage container is of the maximum allowed dimensions and the components are sized to fit as follows: wings inserted whole, fuselage fit in with landing gear and propeller attached, and tail inserted attached to the main tube. Its construction will be of wood panelling with steel reinforcements and will have a hatch style opening at the top that can be secured with drawbolts. Special supports will be placed on the sides to hold the wings by the extremities and others will be inserted on the floor to stop the fuselage from rolling during transport. Lastly, a support will hold the tail surfaces and tube securely to avoid damage. The handles will be placed on either side and there will be wheels in the corners. These are included in size limits as specified by the regulations.

6.4 Innovation & Sustainability

The choice of lamination and 3D printing techniques is largely driven by their ability to facilitate hands-on experimentation with innovative methods and serve as a testing ground for experimental techniques developed by external companies. One of the main challenges in composite manufacturing is its environmental impact, as both composite lamination and mould production are known to be resource-intensive and wasteful. To improve the sustainability of the O.L.I.V.I.A. project and future developments, the team partnered with a sponsor company involved in an initiative called ZeroWasteLIFE. This project aims to address sustainability issues in composite manufacturing by actively reducing CO emissions, optimizing energy consumption, minimizing landfill waste, and contributing to a healthier environment. As part of this effort, the wing moulds were designed to minimize material usage and allow for recycling in future manufacturing processes. Additionally, large composite material suppliers frequently dispose of substantial amounts of recently expired yet fully usable materials. To reduce waste, all composite materials used in O.L.I.V.I.A. were sourced from surplus stock that would have otherwise been discarded. Research is also ongoing into eco-friendly resins, which could be applied in future projects.

The lighter weight of 3D-printed moulds enables further experimentation with advanced techniques, such as closed-mould autoclave curing, as described in the Lamination section. Additionally, ongoing tests are being conducted with various 3D printing filaments to explore ways to independently produce small moulds, further increasing flexibility and sustainability in the manufacturing process.

7 Qualification Test Plan

O.L.I.V.I.A. project's Qualification Test Plan is summarised in the table below.

ID	Objective	Method	Success Criteria	Results and Date
T1	Battery Endurance (20% remaining after 15 min)	Bench test simulating flight power requirements (derived from equilibrium equations and CFD polar data), followed by flight tests.	Battery must retain at least 20% charge after a 15-minute mission.	Completed successfully - awaiting flight testing.
T2	Propeller Static Tests	Measure maximum available thrust during takeoff using a test bench.	Thrust meets required value for take-off feasibility.	Completed successfully (<i>APC 15x10E</i>).
T3	Flight Termination System (FTS)	Simulate a competition-like mission with the drone held on the ground to ensure that breaching the fly-zone triggers the FTS.	FTS must activate correctly upon fly-zone breach.	Tests on Fabia drone - expected 04/04.
T4	Radio Link Range (≥ 1 km)	Field test separating components by at least 1.5 km to verify correct operation.	Reliable communication at ≥ 1 km range.	Scheduled for testing - expected 04/04.
T5	Flight Envelope Limits (Max 200 AGL, 40 kts, 30° bank)	Tune autopilot parameters to impose these limits and verify correct implementation through flight tests.	Flight limits enforced successfully.	Tested on Fabia - expected 04/04.
T6	Payload Release System	Step 1: Static test, releasing the payload while the drone is held. Step 2: In-flight test to verify correct drop mechanism and accuracy.	Payload release is safe, reliable, and accurate.	Awaiting fuselage manufacture - 30/04.
T7	MTOM $\leq 10.0kg$	Weighing scales – aircraft fully loaded and with dummy weighted tracker.	MTOM $\leq 10.0kg$.	Awaiting manufacture – expected 07/05.
T8	Weight, Balance, and CG Location	Use calibrated balances to verify the total weight and CG location.	CG within safe limits for stability.	Awaiting manufacture - expected 07/05.
T9	Structure static integrity	The aircraft will be fully loaded and a structural integrity assessment will be performed.	Expected deformations of the wing and fuselage.	Awaiting manufacturing - expected 07/05.
T10	5-minute assembly of UAS and GCS	The aircraft and GCS will be assembled and prepared for flight, simulating the fly-off event.	Compliance with competition regulation and safety operations ensured.	Awaiting flight testing - 01/06.
T11	Static trimming	Aircraft trimmed both on-ground and during human-piloted flight.	Aircraft has successfully been trimmed.	Awaiting flight testing - expected 01/06.
T12	Successful no-payload mission scenario	Full fly-off-like mission will be simulated without loading the payload.	Mission scenario completed ensuring all competition requirements and safe operation procedures.	Awaiting flight testing - expected 07/06.
T13	Successful payload mission scenario	Full fly-off-like mission will be simulated, including the payload release procedure.	Mission scenario completed ensuring all competition requirements and safe operation procedures.	Awaiting flight testing - expected 07/06.
T14	10 m take-off and 20 m landing	The aircraft will demonstrate its STOL capabilities in fly-off-like conditions.	Compliance with competition rules and safety operations.	Awaiting flight testing - expected 10/06.

Table 11: Test Plan and Safety Verification.

8 Cost Breakdown

Expenses incurred for the project are here presented in a table divided into two sections. The first section lists the costs of individual components, including both their estimated market value at the beginning of the project (**Estimated Value**) and the actual cost incurred (**Actual Value**). This provides insight into the total expenditures and their alignment with initial projections. The second section details the costs associated with manufacturing processes and equipment. It is important to note that, unlike individual components, accurately estimating the cost of some of these processes—entirely handled by sponsors—is challenging, as they involve expenses that are not disclosed. Annotations have been added alongside these values to clarify their assessment.

Item	Estimated Value	Actual value	Sponsored portion	COTS
Motor	134.95€	174.39€	0%	yes
Propeller	8€	8€	0%	yes
Battery	72.29€	72.29€	0%	yes
Ogive	5.50€	5.50€	0%	yes
Wheels	20€	not bought yet	0%	yes
Servos	528€	264€	50%	yes
ESC	134.95€	174.39€	0%	yes
Flight Controller	104.99€	106.99€	0%	yes
Receiver (radio)	17.91€	23.53€	0%	yes
Receiver (ground)	60.57€	60.57€	0%	yes
GPS compass	75.98€	76.50€	0%	yes
Lidar	41.69€	41.19€	0%	yes
Airspeed sensor	57.39€	57.39€	0%	yes
Micro SD	0.93€	2.79€	0%	yes
Cables and switches	60€	not bought yet	0% (estimated)	yes
Composite materials	1450€	290€	80%	no
Tubes	100€	not bought yet	0-100%	no
3D printing material	100.91€	45.36€	55,05%	no
Production and Testing				
Equipment	1134.95€	1134.95€	0%	no
Molds materials	2800€	2450€	12,5%	no
Molds printing time	10840€	0€	100%	no
Composite lamination and curing	11600€ ^a	0€	100%	no
Test flights	360€	60€ ^b	0%	no

Table 12: Cost Breakdown.

^aIt accounts only for autoclave hours, estimating €400 per hour for a total of 29 hours (7 for the wings, 7 for the fuselage, 7 for the control surfaces, 4 for the honeycomb plate, and 4 for other parts). It does not include the materials used at the company (vacuum bags, valves, barometers, etc. and other related expenses).

^bEstimate as of today, the testing campaign is not yet complete.

Total expenses are summarized in the table below in both euros and pounds.

	Estimated Value [€]	Actual value [€]	Estimated Value [£]	Actual value [£]
Components	2974	1583	2489	1330
Production	26 735	3645	22 378	3051
Total	29 709	5228	24 867	5196
Off the shelf	1323	1148	1108	964

Table 13: Total Expenses.